

STAT 579: Applied Causal Inference

Tentative Syllabus Syllabus

Fall 2025

Course: STAT 579 Applied Causal Inference

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Class Meetings: Tuesdays and Thursdays, 9:30 AM - 10:45 AM

Course Overview

This graduate-level course provides a comprehensive introduction to the theory and application of causal inference methods, with a strong emphasis on biostatistical applications. Moving beyond mere association, students will learn formal frameworks for defining and estimating causal effects, particularly from observational data common in public health and clinical research. The course will delve into foundational concepts like potential outcomes and causal graphs (DAGs), explore key challenges such as confounding and selection bias, and cover essential methods including propensity scores, instrumental variables, mediation analysis, and techniques for time-varying treatments. Through a blend of theoretical lectures, applied examples, R assignments completed independently, and critical discussion of research papers, students will develop not only the technical skills to apply these methods but also the critical thinking abilities necessary to rigorously evaluate causal claims in scientific literature and public discourse.

My Approach to Causal Inference: This course emphasizes that causal inference is not merely a toolkit of statistical methods, but a way of thinking rigorously about cause and effect. We will constantly return to foundational principles: clearly defining the causal question and estimand, making all assumptions explicit (often with Directed Acyclic Graphs - DAGs), understanding what each method can and cannot do under those assumptions, and critically assessing the robustness of our conclusions. Expect to be challenged to think deeply and defend your reasoning!

Course Goals and Objectives

Course Goals

- To equip students with a strong theoretical foundation in the principles of causal inference.
- To enable students to apply a range of modern causal inference methods to real-world biostatistical problems.
- To foster critical thinking skills necessary to identify causal questions, recognize threats to valid causal inference, and evaluate causal evidence.
- To develop practical skills in using statistical software for causal data analysis.

Learning Objectives

Upon successful completion of this course, students will be able to:

- Distinguish between association and causation and formally define causal effects using the potential outcomes framework.
- Represent causal structures using Directed Acyclic Graphs (DAGs) and use them to identify confounding, selection bias, and paths to block for identification.
- Identify, articulate, and critically evaluate the assumptions required for causal identification under different scenarios (e.g., conditional exchangeability, positivity, SUTVA, instrumental variable assumptions).
- Apply propensity score methods (matching, stratification, inverse probability weighting) to estimate causal effects in observational studies, including assessing their plausibility and checking diagnostics like balance and overlap.
- Implement g-computation and inverse probability weighting for estimating causal effects of time-varying treatments using Marginal Structural Models, clearly articulating the required sequential assumptions.
- Understand the principles, stringent assumptions, and potential pitfalls of Instrumental Variable analysis and apply relevant methods.
- Formulate and assess causal mediation questions and apply methods to decompose effects into natural direct and indirect effects, understanding the strong no-unmeasured-confounding assumptions required.
- Conduct sensitivity analyses to evaluate the robustness of causal estimates to violations of key assumptions (e.g., unmeasured confounding).
- Critically evaluate causal claims in published biostatistical and public health research.
- Implement causal inference methods using R statistical software, including relevant packages, and interpret the output in a causal context, grounded in the method's assumptions.

Learning Outcomes

These outcomes are directly derived from the learning objectives and are stated in measurable terms:

- Students will be able to draw and interpret DAGs for given causal scenarios, identifying key structural features like confounders, colliders, and mediators.
- Students will be able to state the Stable Unit Treatment Value Assumption (SUTVA) and discuss its implications and potential violations in various study designs.
- Students will be able to apply the backdoor criterion using a DAG to identify sufficient sets of confounders to adjust for, and explain why a given set is or is not sufficient.
- Students will be able to implement and interpret propensity score matching/weighting using R, including assessment of covariate balance and overlap.
- Students will be able to fit and interpret Marginal Structural Models for time-varying treatments using IPW in R, clearly stating the assumptions for identification of the desired causal parameters.
- Students will be able to apply two-stage least squares (2SLS) for instrumental variable estimation in simple cases, articulate the specific causal effect being estimated (*LATE*), and detail the conditions under which it is identified.
- Students will be able to calculate natural direct and indirect effects using appropriate mediation formulas or regression-based approaches, and state the key assumptions for their identification (e.g.,

cross-world counterfactual independence assumptions).

- Students will be able to perform basic sensitivity analyses for unmeasured confounding using statistical software and explain the implications of the findings for the robustness of the primary causal conclusions.
- Students will be able to critique the causal inference methods used in a peer-reviewed journal article, focusing on the alignment of the research question, data, methods, assumptions, and conclusions.
- Students will be able to write R code to perform causal analyses on provided datasets and interpret the output, distinguishing between associational and causal interpretations based on the methods and assumptions.

Assignments and Assessments

Assignments and assessments are designed to reinforce theoretical understanding and develop practical application skills, directly aligning with the learning outcomes.

- **Weekly Problem Sets (20%):** Short problem sets focusing on theoretical concepts, DAG interpretation, and small calculation exercises. These will often require students to articulate assumptions for specific scenarios and justify methodological choices.
- **Applied R Assignments (25%):** Students will complete R coding assignments applying methods covered in lectures to synthetic or real datasets. These assignments are designed to reinforce practical implementation skills and will be completed independently. Detailed instructions and R code examples from lectures will be provided, and support will be available through instructor office hours.
 - **R Assignment 1 (associated with Week 3-5 concepts, due around Week 6/7):** DAGs, Simulation, Bias, and introduction to G-formula concepts. Using `dagitty` to represent causal structures. Simulating data (e.g., with confounding, mediation, colliders) to demonstrate the impact of these structures on naive associations versus true causal effects.
 - **R Assignment 2 (associated with Week 6 concepts, due around Week 8/9):** Propensity Score Methods. Implementing propensity score matching (`MatchIt`) and weighting (`WeightIt`), including propensity score estimation, checking covariate balance (`cobalt`), and assessing overlap.
 - **R Assignment 3 (associated with Week 8 concepts, due around Week 10/11):** Instrumental Variable Analysis. Implementing an IV analysis (`ivreg`), assessing instrument strength, and articulating IV assumptions in the context of the data.
 - **R Assignment 4 (associated with Weeks 9-10 concepts, due around Week 11/12):** Causal Mediation Analysis. Implementing a causal mediation analysis (`mediation` package) based on the two weeks of material, interpreting direct and indirect effects, and discussing the strong assumptions required.
 - **R Assignment 5 (associated with Week 12 concepts, due around Week 13/Finals Week):** Marginal Structural Models. Implementing a simple MSM using IPW (e.g., with `WeightIt` for weights and `geepack` or `survey` for robust standard errors), clearly stating sequential exchangeability and positivity assumptions.
- **Midterm Exam (20%):** Covers foundational concepts (Potential Outcomes, DAGs, Confounding, Identifiability, Propensity Scores). Mix of conceptual questions, DAG drawing/interpretation, simple calculations, and short explanations of assumptions.
- **Final Project (35%):** A capstone project where students apply causal inference methods to a real-world biostatistical dataset (publicly available, from their own research with approval, or provided).
 - **Project Proposal (due mid-semester, Week 8, 5% of final project grade):** Defining a

clear causal question, identifying a suitable dataset, constructing an initial DAG based on domain knowledge, and outlining potential methods and estimands.

– **Final Report & Presentation (30% of final project grade):**

- * Defining a clear causal question and target estimand.
- * Constructing a well-justified DAG based on background knowledge, explicitly stating the causal assumptions it represents.
- * Choosing and implementing appropriate causal inference methods, justifying the choice based on the DAG, data structure, and research question. This includes verifying assumptions where possible (e.g., positivity, balance).
- * Conducting appropriate sensitivity analyses for key assumptions.
- * Writing a comprehensive report summarizing methods, findings (with appropriate causal language), and discussing limitations and threats to validity.
- * A short (5-7 minute) in-class or recorded presentation of their project.

Recommended Materials

Primary Textbooks:

- Hernán MA, Robins JM. *Causal Inference: What If*. Available online: <https://www.hsph.harvard.edu/miguel-hernan/causal-inference-book/>. **Our primary reference, providing a comprehensive framework unifying potential outcomes, DAGs, and advanced methods, with rich examples from epidemiology and public health.**
- Pearl J, Glymour M, Jewell NP. *Causal Inference in Statistics: A Primer*. Wiley, 2016. **An accessible introduction to Pearl's structural causal model framework, focusing heavily on DAGs and do-calculus for identification. Excellent for conceptual understanding of graphical models.**
- Imbens GW, Rubin DB. *Causal Inference for Statistics, Social, and Biomedical Sciences: An Introduction*. Cambridge University Press, 2015. **A deep dive into the potential outcomes framework, with rigorous treatment of randomized experiments and observational study designs, particularly matching methods. Valuable for its statistical formalism and detailed examples.**

Supplementary Text/Readings:

- VanderWeele TJ. *Explanation in Causal Inference: Methods for Mediation and Interaction*. Oxford University Press, 2015. **The authoritative text for modern causal mediation analysis and understanding effect modification/interaction on causal scales.**
- Brumback BA. *Fundamentals of Causal Inference with R*. CRC Press, 2021. **Given that there are no dedicated lab sessions, students are strongly encouraged to utilize the R examples in Brumback (2021) and other provided resources when working on R assignments.**
- Seminal papers by Rubin (e.g., Rubin, D. B. (1974). Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of Educational Psychology*, 66(5), 688–701), Pearl, Rosenbaum & Rubin (e.g., Rosenbaum, P. R., & Rubin, D. B. (1983). The central role of the propensity score in observational studies for causal effects. *Biometrika*, 70(1), 41–55), Hernán & Robins. Specific papers will be assigned or recommended for relevant weeks.
- Relevant chapters from other biostatistics/epidemiology texts discussing confounding, bias, etc.

Datasets:

- Simulated datasets to illustrate specific biases or method applications.
- Publicly available health datasets (e.g., NHANES subsets, data accompanying textbooks or papers like those from the Framingham Heart Study or specific trial emulations).

Software & Tools:

- **R:** Primary statistical software. (Students should have R and RStudio installed).
- **R Packages:** `dagitty` (or online DAGitty tool: <http://dagitty.net> – we will use this extensively to visualize causal assumptions and identify adjustment sets), `MatchIt`, `WeightIt`, `cobalt`, `marginalEffects` (for obtaining interpretable marginal effects/predictions), `mediation`, `ivreg`, `geepack` (for GEE in MSMs) or `survey` (for weighted analyses), `survival` (for survival outcomes in MSMs), potentially `EValue`, `SenseMakes`, or `sensitivitymw` for sensitivity analyses. General utility: `dplyr` for data manipulation, `ggplot2` for visualization.
- **DAGitty:** Web-based tool (<http://dagitty.net>) or R package for drawing and analyzing DAGs.

Teaching Methods

A blended approach combining theoretical instruction with supported independent application is crucial for this topic.

- **Lectures:** Covering theoretical concepts, introducing methods, and walking through examples. Emphasis on assumptions underlying each method.
- **Flipped Classroom/Pre-recorded Mini-Lectures:** For some foundational topics, assign pre-recorded videos or readings and use class time for Q&A and problem-solving.
- **In-Class Activities:** Short exercises like drawing DAGs for a given scenario, interpreting output from a causal analysis, identifying the target estimand, or debating the plausibility of assumptions.
- **Software Demonstrations:** Live coding sessions in R during lectures to show how to implement methods and, critically, how to check assumptions and interpret outputs. These demonstrations will serve as a primary guide for the Applied R Assignments.
- **Independent Application and Support:** Students will apply concepts through R assignments completed outside of class. Detailed assignment instructions, code examples from lectures, and instructor office hours will provide support for these practical exercises.
- **Paper Discussions:** Critical analysis of assigned research papers focusing on: the precise causal question, the assumed DAG (even if not explicitly drawn by authors), the chosen methods, the stated and unstated assumptions, appropriateness of methods for assumptions, potential for unmeasured confounding or other biases, and the justification for the causal conclusions drawn.
- **Case Studies:** Work through detailed examples of causal inference applications in specific biostatistical contexts.

Prerequisites

Students should have a solid foundation in core statistical and biostatistical concepts.

- **Statistics:** Graduate-level probability and statistical inference. Familiarity with linear regression, logistic regression, and basic generalized linear models. Understanding of concepts like confounding and interaction in a traditional statistical modeling context.
- **Probability:** Basic probability theory, conditional probability, Bayes' theorem.

- **Statistical Software:** Intermediate proficiency in R. This includes data manipulation (e.g., using `dplyr` or base R equivalents), fitting regression models (`lm`, `glm`), basic plotting (`ggplot2` or base R), and writing simple functions or scripts. The course R assignments will primarily use R.

Backward Design Alignment

The course structure is explicitly designed to build towards the stated learning outcomes and goals, culminating in the final project where students synthesize their knowledge.

- **Outcomes shape Content:** The topics covered in the weekly syllabus (DAGs, Potential Outcomes, Propensity Scores, etc.) are selected specifically to address the learning outcomes.
- **Outcomes shape Activities:** In-class activities and R assignments are designed to provide hands-on practice with the skills listed in the outcomes (drawing DAGs, implementing methods in R, interpreting results).
- **Outcomes shape Assessments:** Problem sets test foundational understanding. R assignments test practical implementation. The midterm assesses understanding of core frameworks and methods. The final project requires students to demonstrate mastery of multiple outcomes by applying methods to a real problem and critically evaluating their findings. For example, the ability to “implement propensity score matching/weighting using R” (an outcome) is directly assessed in an R assignment and potentially utilized in the final project. The ability to “critique causal inference methods” is addressed in paper discussions and assessed in the final project report.

Detailed Week-by-Week Syllabus (13 Weeks + Final Project)

This structure provides a logical flow, building from fundamental concepts to more advanced methods and applications. Each week will involve discussion of the assumptions critical to the methods presented. Assignments involving R will be completed independently, supported by in-class demonstrations and office hours.

Week 1: Introduction to Causal Inference

- The importance of causal questions in science, especially biostatistics.
- Association vs. Causation: “Correlation does not imply causation” – why and when?
- Examples of causal pitfalls (e.g., Simpson’s Paradox, Lord’s Paradox).
- Introduction to the two main frameworks: Potential Outcomes and Causal Graphs (DAGs).
- Syllabus review and course expectations.
- *Readings:* Hernán & Robins (HR) Ch 1; Pearl, Glymour, Jewell (PGJ) Ch 1; Imbens & Rubin (IR) Ch 1.
- *Activity:* Discuss examples of causal claims in the news or research, sketch initial, informal DAGs representing these claims, and identify the desired intervention and outcome.

Week 2: Foundational Concepts: Probability, Statistics, and the Potential Outcomes Framework

- **Review of Essential Probability:** Conditional probability, Bayes’ theorem, expectation, variance, law of total expectation/probability. How these underpin causal concepts.
- **Review of Essential Statistics:** Measures of association (RR, OR, RD), confounding and interaction in standard regression models (`lm`, `glm`), interpretation of regression coefficients. Limitations for causal interpretation.

- **Transition to Potential Outcomes:** Defining individual and average causal effects (ATE , ATT , ATC). The "causal estimand."
- The Fundamental Problem of Causal Inference.
- Stable Unit Treatment Value Assumption (SUTVA): No interference, well-defined interventions. Consistency assumption.
- Randomized experiments as the gold standard; how they address the fundamental problem. Estimating causal effects in ideal RCTs.
- *Readings:* Relevant sections from prerequisite statistics texts (as refreshers); HR Ch 2, 3; IR Ch 2, 3.
- *Activity:* Linking statistical concepts of confounding to causal confounding. Discuss SUTVA violations.

Week 3: Introduction to Causal Graphs (DAGs)

- Nodes, edges (directed, undirected), and paths in DAGs.
- Representing causal assumptions with DAGs; DAGs as non-parametric structural equation models.
- Causal pathways and spurious (non-causal) pathways. D-separation.
- Confounding: Common Causes and the Backdoor Criterion.
- *Readings:* PGJ Ch 2; HR Ch 6, 7.
- *Software:* Introduction to **dagitty** (web or R package). Practice drawing and interpreting simple DAGs, identifying open and closed paths.
- *Assignment:* Problem Set 1 (Conceptual POs, basic DAG drawing, identifying confounders via backdoor criterion conceptually).

Week 4: DAGs, Bias, and Identification

- Colliders and M-Bias: Why conditioning on a collider (or its descendant) induces bias.
- Selection Bias as conditioning on a collider. Examples in study design.
- The Frontdoor Criterion (introduction).
- Using DAGs to identify sufficient adjustment sets for confounding.
- *Readings:* PGJ Ch 3, HR Ch 8.
- *Activity:* Draw DAGs for various scenarios (e.g., from described studies) and identify potential biases (confounding, selection) and propose adjustment sets if appropriate, or state why adjustment is not possible/advisable. Discuss untestable assumptions in DAGs.

Week 5: Identifiability, Exchangeability, and Standardization (G-Formula)

- Formalizing identifiability of causal effects.
- Conditional Exchangeability (Ignorability, no unmeasured confounding).
- The link between DAGs (d-separation, backdoor criterion) and Conditional Exchangeability.
- Positivity (Overlap) assumption.
- Introduction to the g-formula (standardization) as a way to achieve conditional exchangeability through outcome modeling. Assumptions for g-formula: conditional exchangeability, positivity, SUTVA, consistency, correct model specification.
- *Readings:* HR Ch 4, 5; PGJ Ch 3.

- *Software*: Simple g-formula calculation examples in R (e.g., using `predict` after `glm` and averaging, or using `marginalEffects`).
- *R Assignment 1 due soon* (DAGs, Simulation, Bias, G-formula concepts).

Week 6: Propensity Score Methods

- The concept of propensity scores (PS): The balancing score.
- The balancing property and how PS help achieve conditional exchangeability.
- Methods using PS: Matching, Stratification, Inverse Probability of Treatment Weighting (IPTW).
- Estimating Average Treatment Effect (*ATE*) and Average Treatment Effect on the Treated (*ATT*) – clearly define these estimands and how methods target them.
- Assumptions for propensity score methods: Conditional Exchangeability, Positivity (Overlap), SUTVA, Consistency, No unmeasured confounding of $X - Y$ given covariates, Correct PS model specification.
- *Readings*: HR Ch 12, 13; IR Ch 13-18; PGJ Ch 4 (section on PS).
- *R Assignment 2 assigned* (Propensity Score Methods - *MatchIt*, *WeightIt*, *cobalt*).

Week 7: Midterm Exam

- Covers Weeks 1-6 material. Focus on concepts, DAGs, assumptions, and interpretation.

Week 8: Instrumental Variables (IV)

- The concept of an instrumental variable: When unmeasured confounding is a major concern.
- Assumptions for valid instruments: (1) Relevance (associated with X), (2) Exclusion Restriction (affects Y only through X – untestable), (3) Exogeneity/Independence (no common causes of Z and Y , independent of potential outcomes $Y(x)$). Discuss the difficulty of finding valid IVs and justifying assumptions.
- Estimating Local Average Treatment Effects (*LATE*) for compliers. Contrast with *ATE/ATT*. Monotonicity assumption.
- Two-Stage Least Squares (2SLS) as an estimation method.
- *Readings*: HR Ch 16; IR Ch 23; PGJ Ch 5.
- *R Assignment 3 assigned* (Instrumental Variable Analysis - *ivreg*).
- *Final Project Proposal due*.

Week 9: Causal Mediation Analysis - Part 1: Concepts and Identification

- Defining direct and indirect effects: Decomposing a total effect. Purpose of mediation analysis.
- Traditional approaches (Baron-Kenny) and their limitations for *causal* mediation.
- Counterfactual definitions: Natural Direct Effects (*NDE*), Natural Indirect Effects (*NIE*), Controlled Direct Effects (*CDE*). Total Effect.
- DAGs for mediation: Visualizing paths and confounding.
- Identification of *NDE* and *NIE*: The four key no-unmeasured-confounding assumptions ($X - Y|M, C$; $X - M|C$; $M - Y|X, C$; and no $M - Y$ confounder affected by $X|C$). Discussion of cross-world counterfactuals and their independence.

- *Readings:* VanderWeele Ch 1-2; PGJ Ch 4 (section on mediation); HR Ch 9 (briefly, for contrast with advanced methods if any). Relevant articles on mediation assumptions.

Week 10: Causal Mediation Analysis - Part 2: Estimation, Interpretation, and Advanced Issues

- Estimation methods based on regression (for continuous outcomes, binary outcomes).
- Difference method vs. Product method (and their assumptions).
- Using software (e.g., `mediation` package in R): Implementation and interpretation of output.
- Interaction (effect modification) in mediation: Does the direct or indirect effect vary by baseline covariates?
- Brief discussion of sensitivity analysis for mediation assumptions (conceptual).
- Limitations and common pitfalls in mediation analysis.
- *Readings:* VanderWeele Ch 3, 4; Relevant documentation for R packages.
- *R Assignment 4 assigned (Causal Mediation Analysis - `mediation` package).*

Week 11: Sensitivity Analysis (General Unmeasured Confounding)

- The importance of assessing sensitivity to unmeasured confounding for primary causal effects (e.g., from PS or G-formula methods).
- Methods for sensitivity analysis:
 - Cornfield conditions / “Bounding factor” approach.
 - E-values: concept, calculation, and interpretation.
 - Rosenbaum bounds for matched studies (if matching is heavily emphasized).
- Interpreting sensitivity analysis results: How strong would unmeasured confounding need to be to alter conclusions?
- *Readings:* HR Ch 14; IR Ch 21; Ding & VanderWeele (2016) on E-values; Rosenbaum (2002) *Observational Studies* (relevant chapters).
- *Activity/Discussion:* Calculating and interpreting E-values for published results or hypothetical scenarios.

Week 12: Causal Inference with Time-Varying Treatments & Marginal Structural Models (MSMs)

- Challenges with time-varying exposures/treatments: Time-varying confounding affected by prior treatment (feedback loops). Why standard regression/PS methods fail.
- Introduction to Marginal Structural Models (MSMs). Target estimand: effect of a treatment strategy or regimen.
- Using Inverse Probability of Treatment (and censoring, if relevant) Weighting (IPW) to fit MSMs. Stabilized weights.
- Assumptions for MSMs: Sequential conditional exchangeability (no unmeasured confounders for treatment at each time point, given past covariates and treatment), sequential positivity, SUTVA, consistency, correct model specification for weights and outcome model.
- *Readings:* HR Ch 17, 18, 21 (MSMs).
- *R Assignment 5 assigned (Marginal Structural Models - `WeightIt`, `geepack`/survey).*

Week 13: Course Review, Project Work, and Responsible Causal Inference

- Review of core concepts, methods, and the unifying role of assumptions in causal inference.
- Ethical considerations in conducting and reporting causal inference studies.
- Responsible communication of causal findings, including uncertainty and limitations.
- Dedicated time for students to work on their final projects and ask questions.
- Discussion of final project report and presentation requirements.

Finals Week:

- Final Project Report Due.
- Final Project Presentations.

Interactive Learning Activities

- **DAG Drawing Sessions:** Students work in pairs or small groups to draw DAGs for presented scenarios, debate implied conditional independencies, and identify minimal sufficient adjustment sets.
- **Assumption Debates:** Present a study design with a clear causal question and a proposed DAG. Have groups debate whether key causal assumptions (e.g., ignorability, SUTVA, exclusion restriction) are likely to hold and why, and what could be done to improve the design or analysis if assumptions are violated.
- **Instructor-led Code-Along Sessions:** Instructor live codes an example analysis in R during lecture, with students following along and asking questions about interpretation and assumption checking at each step.
- **“Find the Bias” Exercises:** Provide a description of a study and ask students to identify potential sources of confounding, selection bias, or other threats using causal reasoning and explicitly drawing a DAG.
- **Critiquing Published Literature:** Assign a research paper using a causal inference method and have students prepare a structured critique focusing on the causal question, assumptions, methods, and conclusions. Discuss critiques in class. Students should attempt to draw the DAG implied by the authors’ reasoning.
- **Dataset Exploration:** Provide a new dataset and ask students to brainstorm potential causal questions, sketch initial DAGs reflecting plausible relationships, and discuss appropriate methods and their assumptions.
- **“What’s the Estimand?” Exercises:** Given a research question and a method, students identify the precise causal estimand (e.g., *ATE*, *ATT*, *LATE*, *NDE/NIE*) being targeted and under what assumptions.

Assessment Strategies

A mix of formative and summative assessments is used to provide feedback and evaluate learning comprehensively.

- **Formative:**
 - Weekly Problem Sets: Low stakes, designed for practice and feedback on understanding core concepts, calculations, and assumption articulation.
 - In-Class Activities: Informal checks of understanding and engagement.

- Applied R Assignments (initial assignments): Focus on developing practical skills, with feedback on code, interpretation, and assumption checking.
- **Summative:**
 - Midterm Exam: Evaluates understanding of foundational theoretical concepts, basic applications, and the role of assumptions.
 - Applied R Assignments (later assignments): Assess proficiency in implementing specific methods and interpreting results.
 - Final Project (including proposal and presentation): Comprehensive assessment of the ability to define a causal question, choose and implement appropriate methods, conduct sensitivity analysis, and interpret/communicate findings in a real-world context. This is the primary assessment of the applied learning outcomes.
- **Grading Rubrics:** Clear rubrics will be provided for all assignments and the final project to ensure transparency in evaluation criteria, focusing on correct application of methods, clear articulation and justification of assumptions, appropriate interpretation in a causal framework, recognition of limitations, and effective communication.

Software Integration

Software is integrated throughout the course as a tool for applying theoretical concepts and performing analyses.

- **Early Introduction:** Introduce R and relevant packages (like `dagitty`) early in the course during lectures.
- **Applied R Assignments:** Structured R assignments provide hands-on practice with specific methods (propensity scores, IV, mediation, MSMs), completed independently by students.
- **In-Lecture Demonstrations:** Use live coding demonstrations during lectures to illustrate how methods are implemented and what the output looks like, always linking back to the estimand and assumptions. These demonstrations will serve as a key guide for the R assignments.
- **Problem Sets:** May include questions that require students to use R to analyze a small dataset or simulate data to understand a concept.
- **Final Project:** Requires students to independently apply software to a chosen dataset.
- **Focus on Interpretation and Support:** While coding skills are necessary, emphasis will be placed on correctly interpreting the software output in the context of the causal question, the chosen estimand, and the underlying assumptions. Students will be expected to justify why a particular method and its implementation are appropriate for the causal question and assumed DAG, and to critically assess model diagnostics (e.g., balance, overlap). Support for R-related questions will be available during instructor office hours.